

Exam Date & Time: 17-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Number Theory and Cryptography [IT_3130]

Marks: 50

Duration: 180 mins.

PART-B

Answer all the questions.

- 1)
 - a. How many 3-digit non-negative numbers are there that have remainder 7 when divided by 101? Here we assume that 1-digit and 2-digit numbers are also 3-digit, they just start with 0. (5)
 - A) b. Show that the inverse of 5 modulo 101 is 5^{99} .
 - B) Consider an apple which costs **22 dollars**. A customer wants to buy one apple but has only **3-dollar coins** (in any quantity). The shopkeeper has only **5-dollar coins** for giving change. How can the transaction be completed so that both the customer and the shopkeeper are satisfied? Apply suitable logic and compute all possible solutions. (5)
- 2)
 - a. If p and q are distinct prime numbers, prove that $\phi(p \cdot q) = (p-1)(q-1)$ and verify your result for $p=37$ and $q=71$ (5)
 - b. Solve for the positive integer(s) n such that $\phi(n^2) = 1000 \phi(n)$. (5)
 - A)
 - B) a. The first 12 digits of a book's ISBN are 978030640615. Compute & verify the check digit using $d_{13} = (10 - (S \bmod 10)) \bmod 10$ where S is defined as

$$S = \sum_{i=1}^{12} w_i d_i \quad \text{where } w_i = \begin{cases} 1 & \text{if } i \text{ is odd} \\ 3 & \text{if } i \text{ is even} \end{cases} \quad (5)$$

- b. Determine x for the congruence relation, $5x = 31 \pmod{101}$.
- 3)
 - Compute the following
 - (a) Calculate $13^{2010} \pmod{71}$.
 - (b) Calculate $100^{-1} \pmod{2011}$. (5)
 - (c) Calculate $\phi(2010)$.
 - A)
 - B) Let $(G, +)$ be an abelian group. In G we define the product $ab = 0$ for all $a, b \in G$. Show that G is a ring. (5)
- 4)
 - Assume that in hill cipher, we know that the plaintext of our ciphertext message that begins WBVE is INMA compute the Key. (5)
 - A)

- B) The cipher text ALXWU VADCOJO has been enciphered with the cipher
 $C1 = 4P1 + 11P2 \pmod{26}$
 $C2 = 3P1 + 8P2 \pmod{26}$ (5)
- 5) Assume that in round 4 of a DES process for a message, $L4 = 4de5635d$ (in hex), $R4 = 3412a90e$ (in hex) and $k5 = be11427e6ac2$ (in hex). (5)
- A) (a) Find $R5$
 (b) Find $L5$
- B) Analyse how the **key size** and **number of rounds** affect resistance against brute-force and differential cryptanalysis. Would increasing the number of rounds linearly increase security? Justify with mathematical illustration. (5)

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Event Name	Subject
058-END SEMSTER-ODD-2025	IT_3130:Number Theory and Cryptography

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	20
3	3	CO 3	10
4	4	CO 4	10
Total			50

* NA - CO Not Selected

Event Name	Subject
058-END SEMSTER-ODD-2025	IT_3130:Number Theory and Cryptography

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	5	25(50%)
4	Analysis	5	25(50%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)